

SQL Assignment: Farmers Insurance Analysis (PMFBY)

Assignment ID: SQL/02 - Executive Summary

Group: upgrad

Problem and Approach

PMFBY is the central crop insurance scheme. Using NDAP data for 2018 to 2021 we looked at who is covered, what is paid as premium and how this changes over time. The 1,870 row CSV was loaded into MySQL and the 29 questions in the starter file were answered.

Key Insights

Coverage is very concentrated. MP, Maharashtra, UP, Rajasthan and Tamil Nadu account for about 70 percent of farmers in the data. Bid (MH), Ujjain (MP) and Latur (MH) lead on cumulative premium.

Enrolment fell from about 1.5 crore in 2018 to 1.07 crore in 2021, while gross premium grew from Rs. 73 cr to Rs. 131 cr. Fewer farmers but costlier policies.

The cost has moved towards the government. The farmer's share of gross premium dropped from 19 percent in 2018 to about 11 percent in 2021. State and central together now fund close to 90 percent.

Female participation stays around 14 to 16 percent in all four years. Small and marginal farmers form 78 to 86 percent of the pool, the audience the scheme is meant to reach.

Assumptions

- Premium columns are in lakhs and InsuredLandArea in thousand hectares (column_description.csv). Thresholds (500K, 20 cr, 100 cr) were converted to raw units in the queries.
- The Year column is loaded as Year_ since Year is reserved in MySQL. srcYear and Year_ are kept in sync.
- Source is one table, so joins use derived subqueries.
- Census population repeats per district, so state population uses max per district then summed.